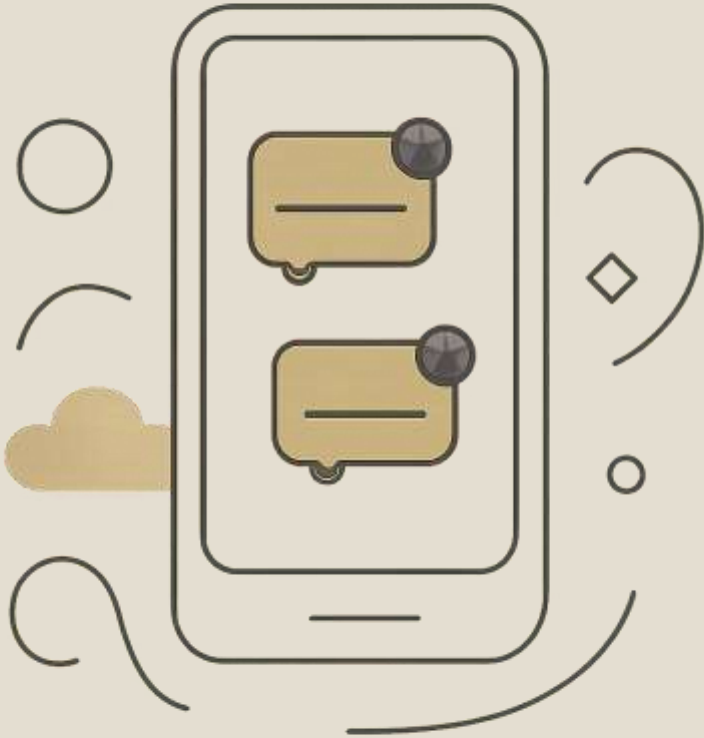




SMS Spam Detection using Naive Bayes

A Machine Learning Based SMS Classification System

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Course: B.Tech Computer Science Engineering (AI & ML)

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Problem Statement & Objective

Problem Statement

- Spam SMS messages often contain advertisements, phishing links, fraudulent offers, or scams.
- Manual filtering is inefficient and unreliable.
- An automated system is required to accurately identify spam messages.

Project Objective

- Develop a Machine Learning model to classify SMS messages as **Spam** or **Ham (Not Spam)**.
- Provide quick, accurate, and reliable predictions through a simple web interface.

Dataset & Text Preprocessing

Dataset

SMS Spam Collection Dataset

Consists of labeled SMS messages:

- **Spam**
- **Ham (Not Spam)**

TF-IDF Vectorization

- Converts processed text into numerical feature vectors.
- Assigns importance to words based on their frequency.
- Enables the Machine Learning model to understand textual data.

Text Preprocessing

Before training the model, each message undergoes the following preprocessing steps:

- Convert text to lowercase
- Remove punctuation and special characters
- Remove stopwords
- Apply Porter Stemming

Machine Learning Model

Algorithm Used

Multinomial Naive Bayes

Why Naive Bayes?

- Fast and computationally efficient.
- Performs exceptionally well for text classification tasks.
- Handles high-dimensional textual data effectively.
- Produces reliable results even with relatively small datasets.

Project Workflow

01

SMS Message

02

Text Preprocessing

03

TF-IDF Vectorization

04

Multinomial Naive Bayes

05

Prediction (Spam / Ham)

Technologies Used & Future Scope

Technologies Used

- Python
- Pandas
- Scikit-learn
- NLTK
- Streamlit
- Joblib

Future Scope

- Improve performance using larger datasets.
- Support multilingual spam detection.
- Implement advanced Deep Learning models such as LSTM or BERT.
- Deploy as a real-time spam filtering application.

Thank You!

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